

Genre Prediction of Movies and TV Series

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Dataset Overview

The dataset contains information about movies and TV shows available across OTT platforms.

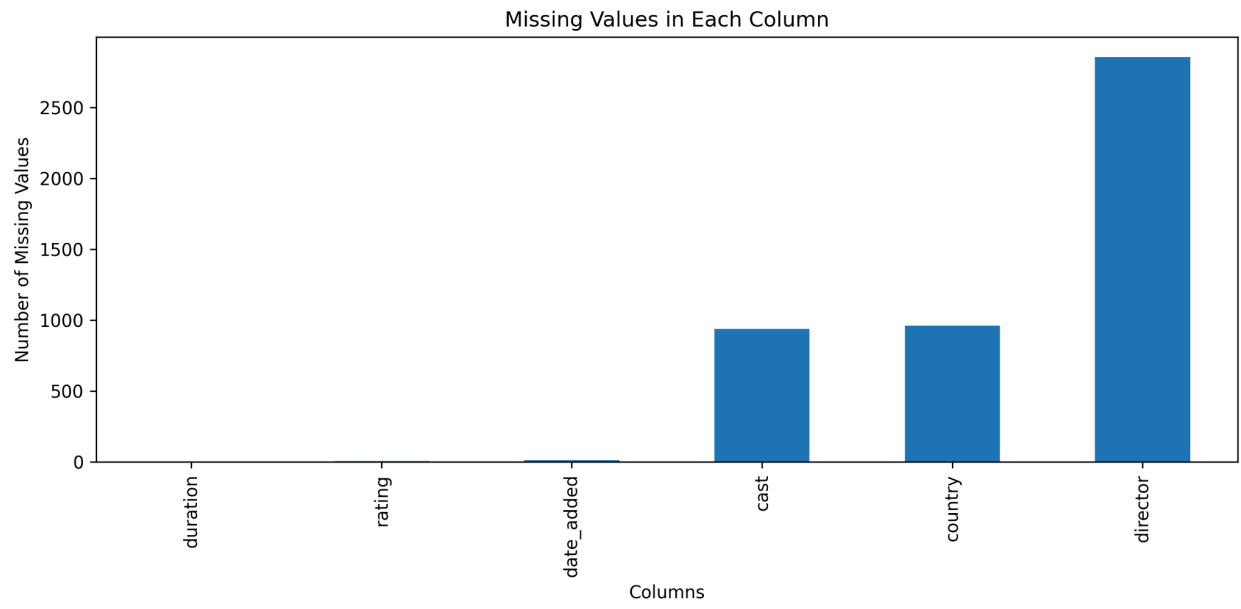
- Total Records: 9,338
- Total Features: 13
- Target Column: listed_in

Key Features

Feature	Description
title	Content Title
description	Description of the content
type	Movie or TV show
rating	Audience rating
country	Country of origin
platform	OTT platform where content is available
listed_in	Genre (Target variable)

Missing Values

Column	Count of Missing values
director	2855
country	962
cast	938
date_added	13
rating	7
duration	3



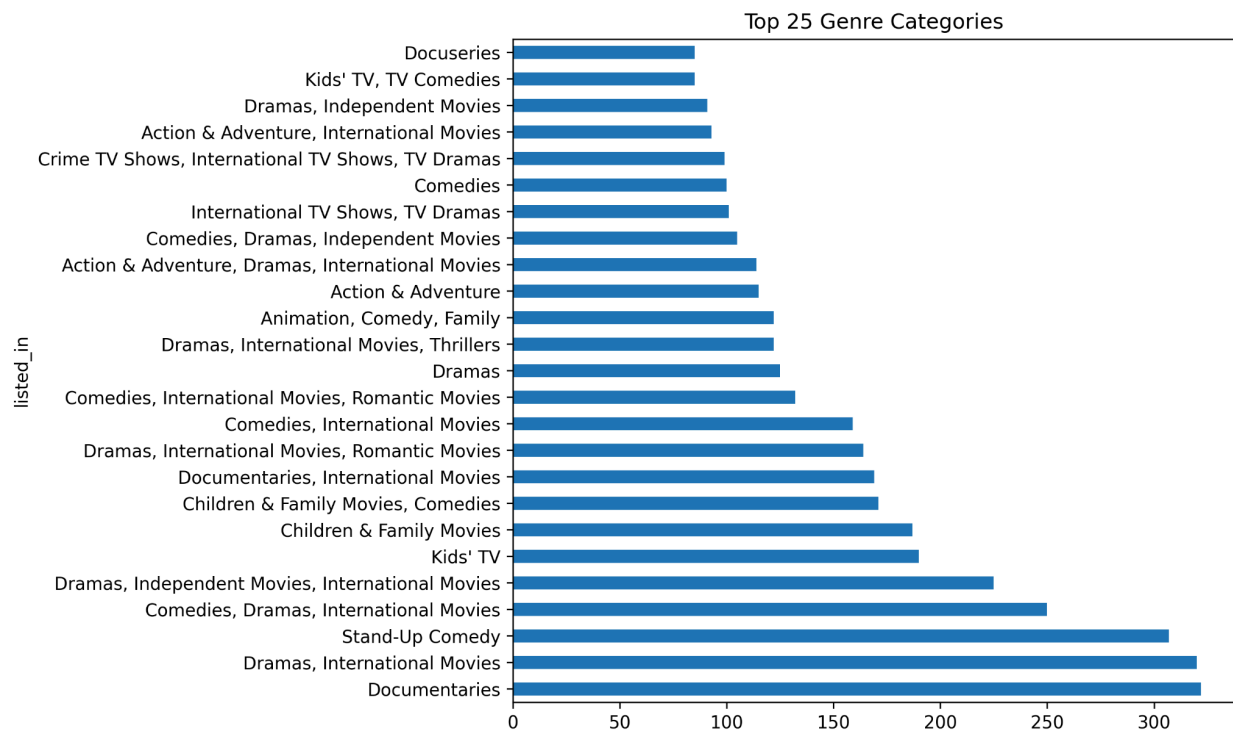
EDA

Genre Distribution

The dataset contained a large number of unique genre combinations, resulting in significant class imbalance.

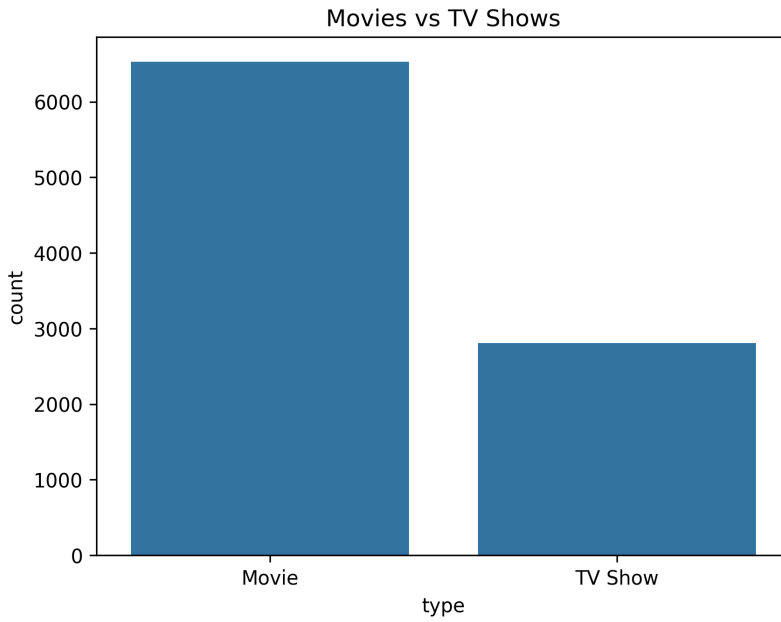
The most frequent genres included:

- Documentaries
- Dramas, International Movies
- Stand-Up Comedy
- Comedies, Dramas, International Movies
- Kids' TV



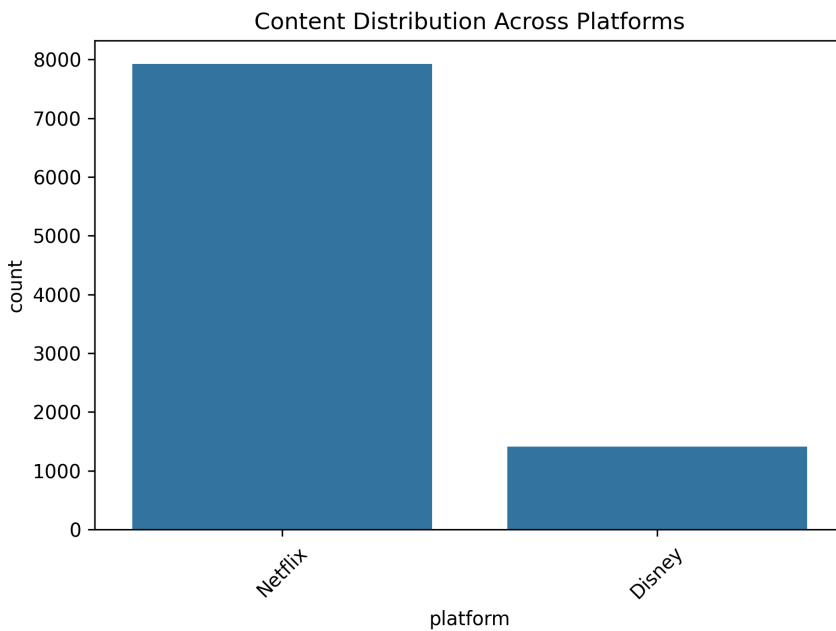
Content Type Distribution

The dataset contains both Movies and TV Shows, with Movies forming the majority of the content.



OTT Platform

The content is distributed within two platforms, Netflix containing the majority of it.



Data Preprocessing

The following preprocessing steps were applied:

1. Removal of infrequent genre categories.
2. Missing categorical values used during modeling were replaced with the value "Unknown".
3. Missing text values were replaced with the value "".
4. Train-test split using an 80:20 ratio.
5. Stratified sampling to preserve genre distribution.

Feature Engineering

Text Features

The content description was transformed using TF-IDF Vectorization.

TF-IDF Parameters:

- Stop words removal: English
- Maximum Features: 25,000
- N-gram Range: (1,2)
- Minimum Document Frequency: 2

Metadata Features

The following metadata fields were incorporated:

- Type
- Rating
- Platform
- Country

Categorical metadata was transformed using One-Hot Encoding.

Combined Feature Space

TF-IDF text features and encoded metadata features were combined into a single sparse feature matrix.

Feature engineering significantly improved model performance compared to using textual descriptions alone.

Model Development

Logistic Regression

A linear classifier is well-suited for high-dimensional, sparse text data.

Parameters:

- `C = 50`
- `max_iter = 5000`

Random Forest

An ensemble tree-based model is used as a non-linear baseline.

Parameters:

- `n_estimators = 100`
- `random_state = 42`

Linear Support Vector Classifier (LinearSVC)

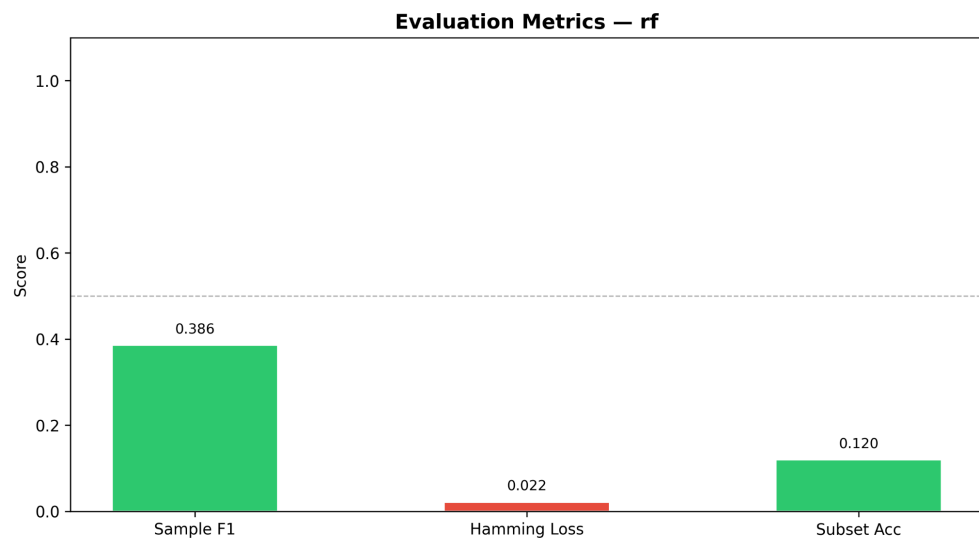
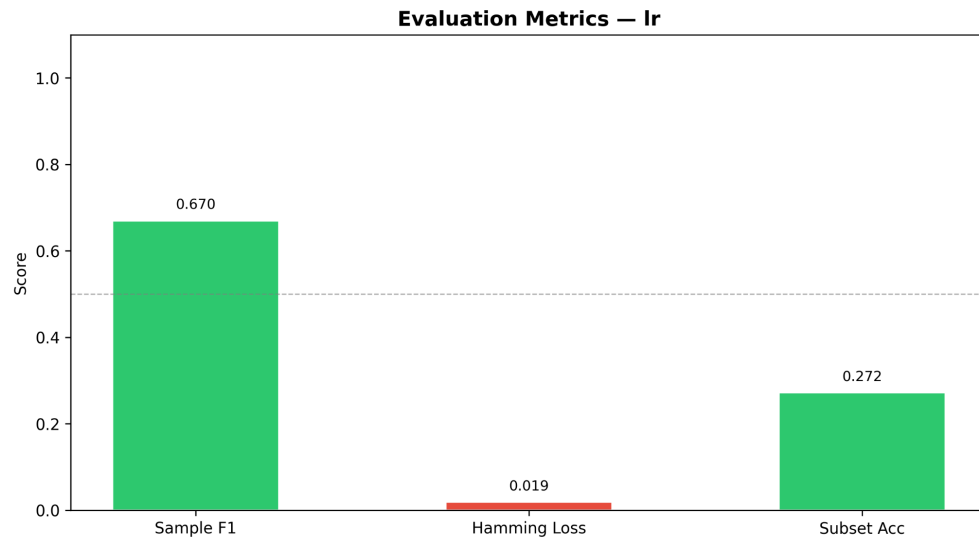
A linear margin-based classifier is commonly used in text classification tasks.

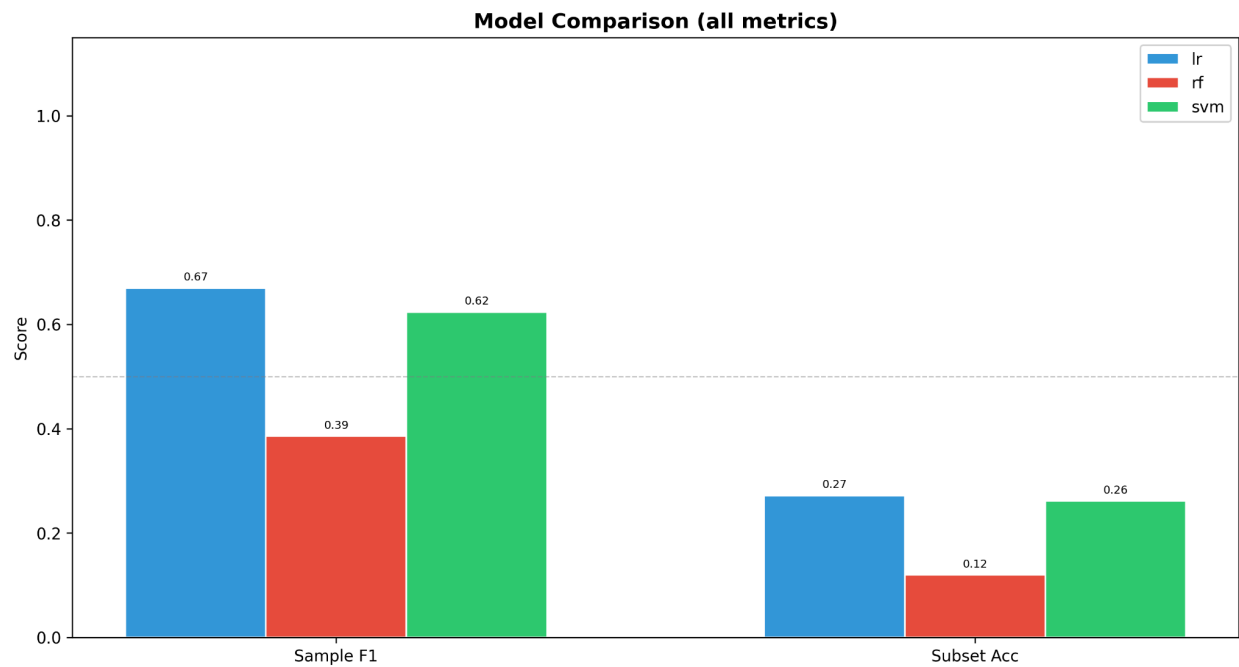
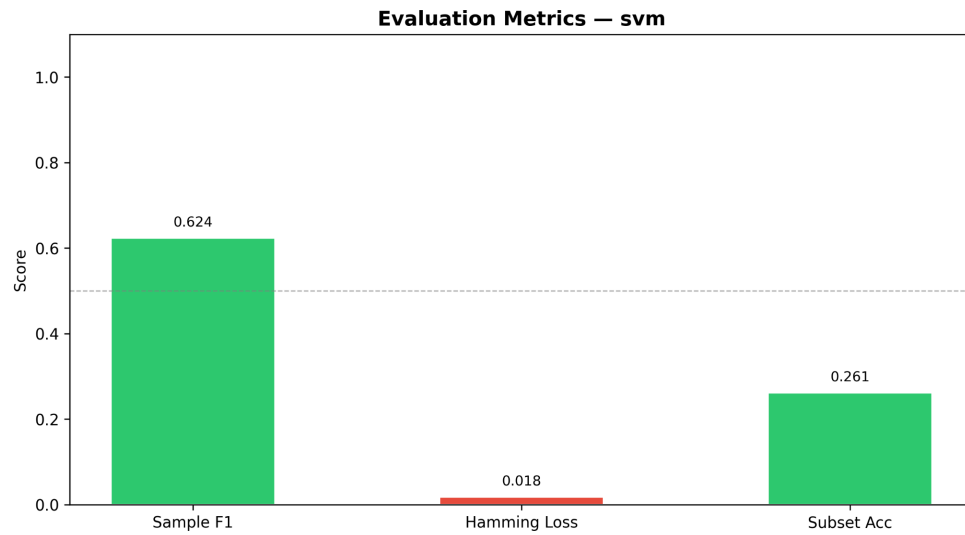
Parameters:

- `C = 50`
- `max_iter = 5000`

Model Evaluation

Models were evaluated using three metrics: Sample F1, Hamming Loss, and Subset Accuracy.





Observations

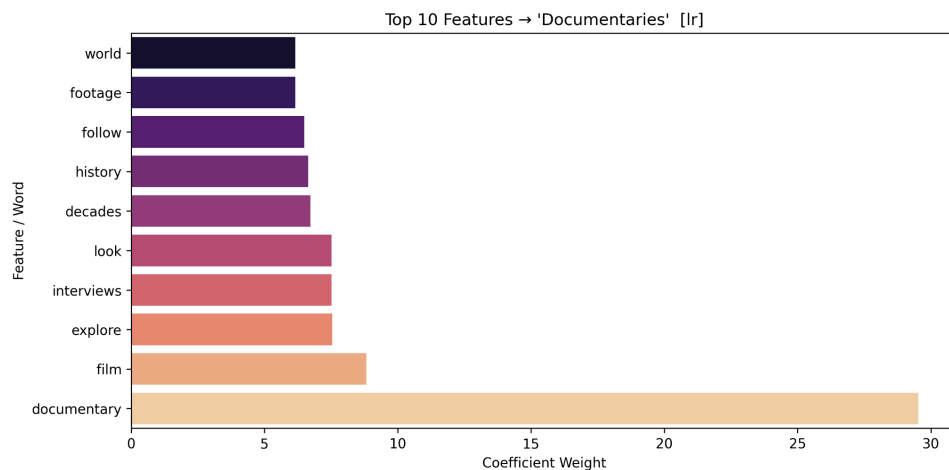
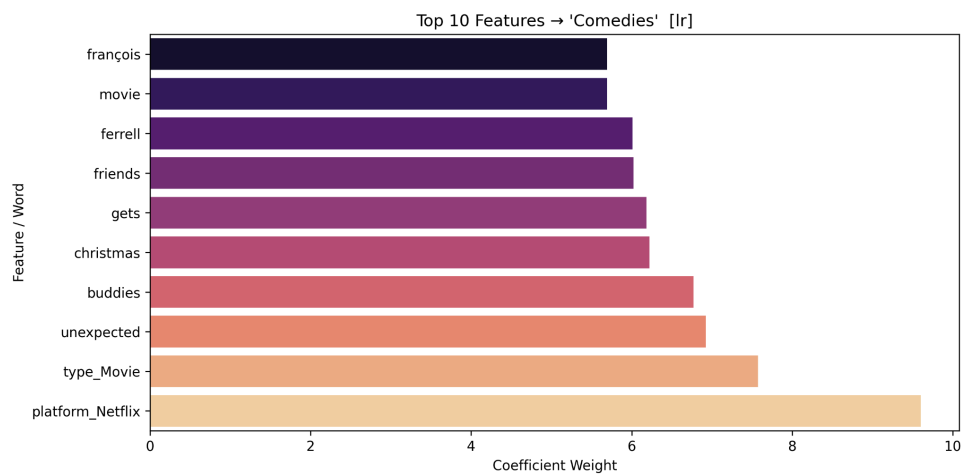
- Logistic Regression achieved the highest overall accuracy.
- LinearSVC produced comparable performance.
- Random Forest performed significantly worse than the linear models.
- Text classification tasks with TF-IDF features tend to favor linear models due to the high dimensionality and sparsity of the feature space.

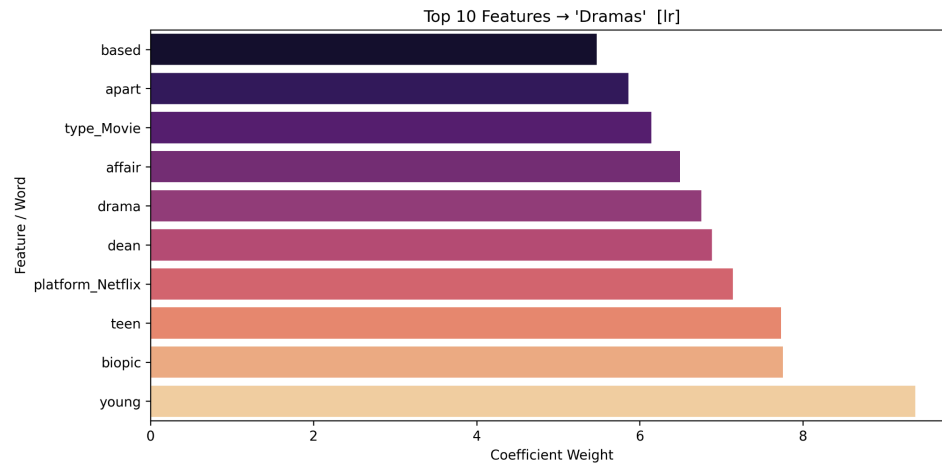
Explainability

Logistic Regression Coefficient Analysis

Because Logistic Regression is a linear model, the learned coefficients can be inspected directly.

For each genre category, the highest positive coefficients were extracted to identify the words and features that most strongly contribute toward that prediction. This analysis provides transparency into how the classifier makes decisions and highlights genre-specific vocabulary patterns.





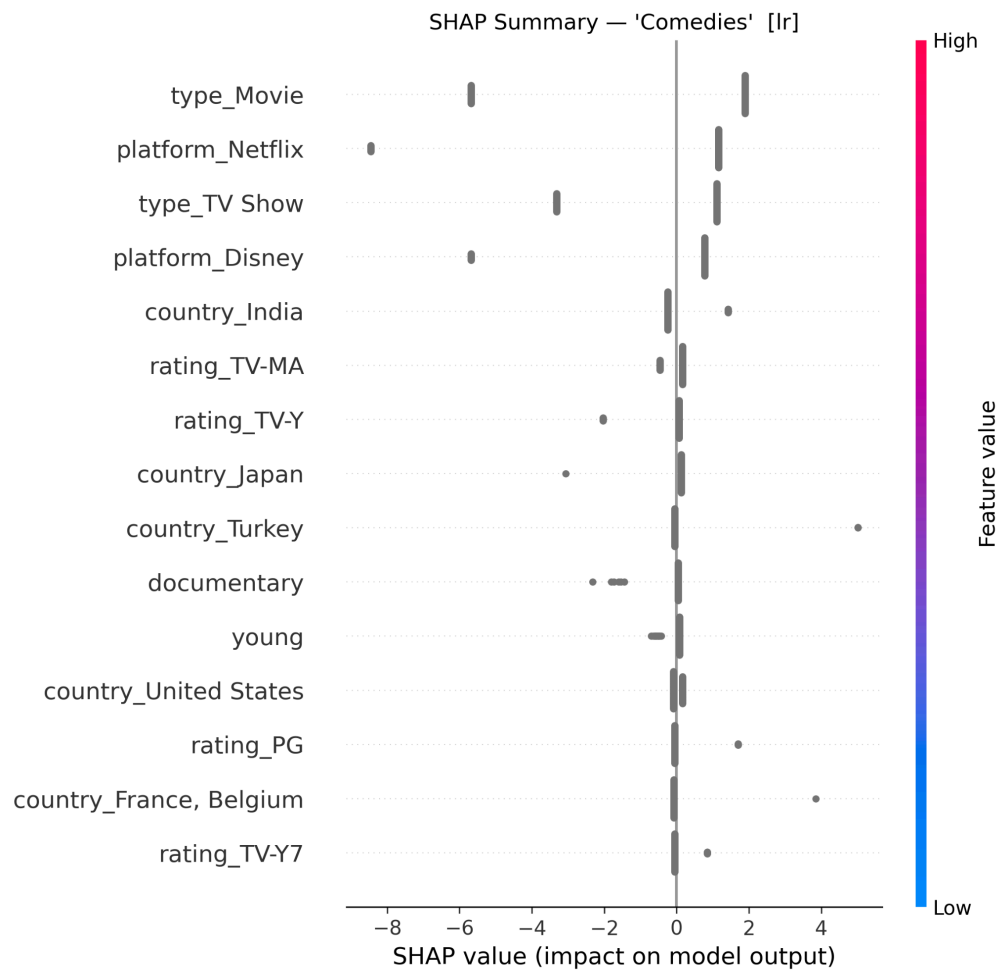
SHAP Analysis

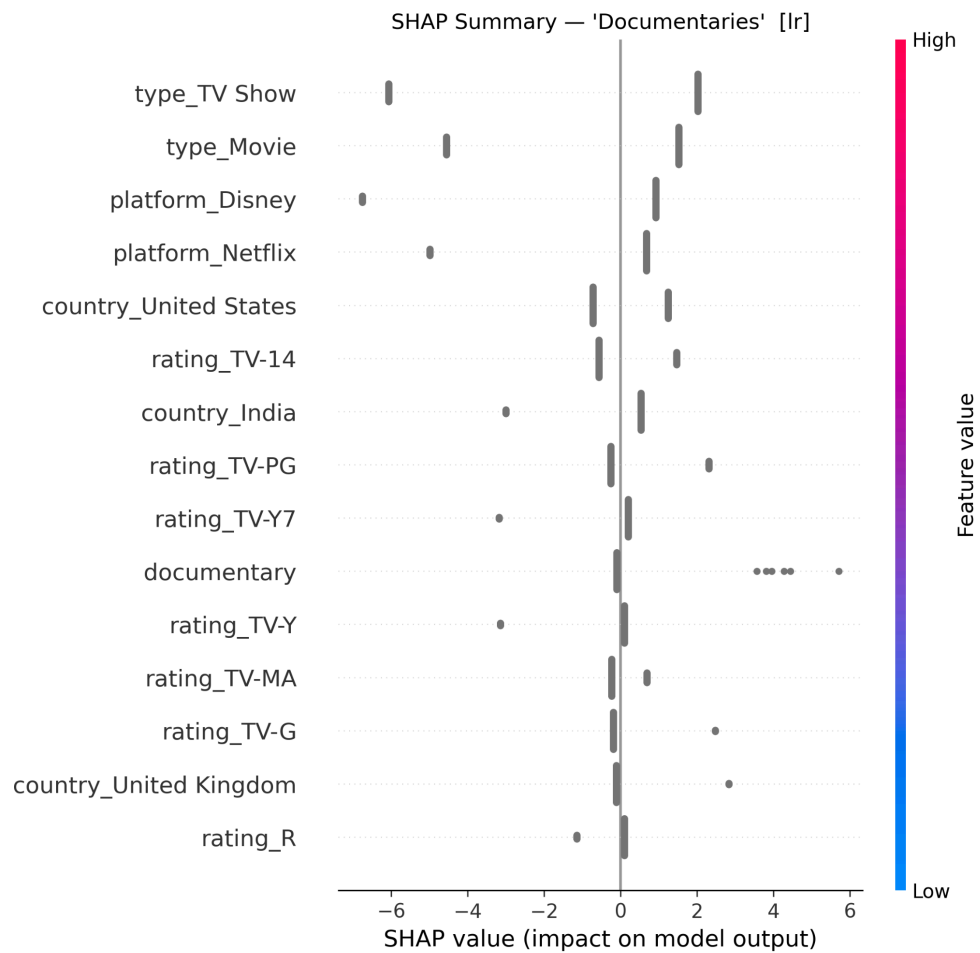
SHAP (SHapley Additive exPlanations) was used through SHAP LinearExplainer to quantify feature contributions.

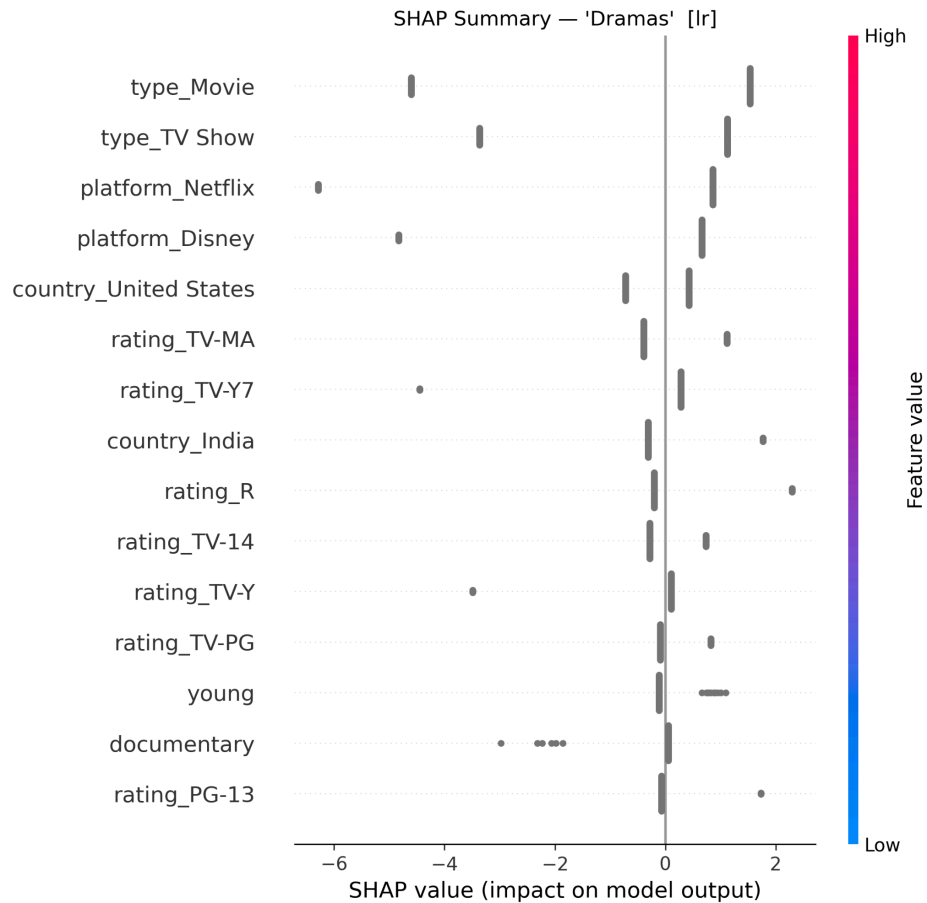
SHAP values provide:

- Feature importance ranking
- Directional impact of features
- Local and global interpretability

Positive SHAP values indicate features pushing predictions toward a genre, while negative values indicate features reducing the likelihood of that genre.







Key Insights

The explainability analysis revealed that:

- Genre-specific keywords strongly influence predictions.
- Metadata features such as content type, platform, rating, and country contribute meaningful predictive signals.
- The model's decisions align with intuitive genre characteristics, increasing confidence in model behavior.

Recommendation

Based on the experimental results, Logistic Regression is recommended as the final model.

Reasons:

1. Highest F1 Score.
2. Fast training and inference.
3. Well-suited for sparse TF-IDF feature spaces.
4. Easier to interpret compared to more complex models.

While LinearSVC achieved comparable performance, Logistic Regression provided slightly better results and was therefore selected as the preferred solution.